(i) Write the nuclear equations corresponding to each of these two decay processes. (2 marks)

*(ii) The half-life of Bi-213 is 46 minutes. At time $t = 0$ s, its number of undecayed nuclei is 8.0×10^{11}. Find its activity (in Bq) at $t = 0$ s. (2 marks)

*(iii) Find the percentage decrease in the activity after 200 minutes. (2 marks)

Answers written in the margins will not be marked.

Figure 12.2(a) Figure 12.2(b)

(i) Compared to placing it in a badge worn on the chest, what is the advantage of placing the chip in the ring worn on the hand ? (1 mark)

(ii) The ring should be worn under the gloves. Explain briefly. (1 mark)

(iii) Which type(s) of radiation (α, β or γ) can reach the chip of the ring ? (1 mark)

END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

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